

Systems Friction as a First-Class Design Requirement for Outsiders in Civic Digital Services

Independent research note (portfolio writeup; not peer-reviewed)

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Abstract

Civic and public digital services routinely assume local context, language fluency, and unwritten process knowledge. For outsiders (non-native speakers, new arrivals, and first-time users), that assumption appears as systems friction: stalled forms, PDF-only “language support,” blaming error messages, and channels that do not match outsider schedules. This independent research note locks a falsifiable claim: treating outsider friction as a first-class design requirement, rather than a post-launch accessibility patch, is a necessary input for accountable civic digital services and public-interest automation. Method: reflective pattern extraction from a Spring 2026 semester abroad in Madrid (Complutense / IU Overseas Study), recorded as a public-safe pattern library in the madrid-ai-ethics repository, plus a reusable Outsider Impact Review one-pager for builders. Contribution at this stage is the claim, five observed patterns, the review artifact, and explicit limits (single observer, one city, reflective method, no controlled trial). Adjacent literature on administrative literacy, administrative burden, and WCAG baselines documents related harms; the operational gap targeted here is a builder-facing pre-automation gate. No quantitative validation is claimed. A minimal dual-review kill test is specified for later work. This PDF is a portfolio-style writeup, not a peer-reviewed publication and not IU or clinic research output.

1 Introduction

Public services increasingly run through forms, apps, and portals. When those systems are designed for people who already know the local script (which office comes first, which document format is accepted, which language completes the task), newcomers experience the same interface as a wall. That wall is easy to misread as personal failure. This note treats it as a design signal.

Situation. Spring 2026, the author lived and studied in Madrid through IU Overseas Study at Universidad Complutense. Coursework supplied vocabulary for institutional design; daily life supplied ground truth for how civic, university, commercial, and transit systems meet someone who does not yet belong. Parallel clinic observation at ServeIT made the same failure modes visible in nonprofit handoffs: tools built for the maintainer’s mental model, not the visitor’s first visit [4].

Gap. Accessibility practice (for example WCAG 2.1 baselines used in clinic work) catches many markup and interaction failures [1]. A happy-path or checklist pass can still miss outsider-specific assumptions about local context, sequencing, and language-as-access. Administrative literacy and administrative burden research documents related costs for migrants and claimants in digital welfare systems [2, 3]. What those bodies of work do not fully supply for student and clinic builders is a

portable pre-ship gate that can stop automation when baseline access fails.

Claims. Stated as claims, not results:

1. Systems friction in civic digital services systematically disadvantages outsiders; the five patterns in Section 4 are the working library from Madrid reflective notes (suggestive, not causal proof).
2. Treating that friction as a first-class design input, via a reusable Outsider Impact Review, is a practice bridge toward more accountable public digital services and public-interest AI adoption.
3. The current method is reflective practice with an explicit kill test; no controlled experiment results are claimed here.

Section 2 situates prior themes; Section 3 describes method and design principles; Section 4 reports observed patterns; Section 5 states limits; Section 6 concludes.

2 Related Work

Three clusters frame the gap. This is not a complete literature survey; deep reading remains owned by the author beyond the sources cited below.

Accessibility and comprehensibility. Clinic practice uses WCAG 2.1 as a baseline for keyboard paths, con-

trast, structure, and related success criteria [1]. Language access and plain-language defaults sit adjacent to comprehensibility even when checklists focus on markup. That overlap is necessary but not sufficient when the primary task is only completable after hidden local prerequisites.

Administrative literacy, burden, and migrant access. Safarov [2] analyzes Russian-speaking migrants navigating digital public services in Finland and revisits administrative literacy as a combination of bureaucratic literacy, structural knowledge, welfare-system navigation, digital skills, and language skills. The work shows that digitalisation can impose rigid requirements that burden or prevent access to entitlements, and that digital skills alone do not explain exclusion. Moynihan, Herd, and Harvey [3] conceptualize administrative burden as learning, psychological, and compliance costs in citizen-state interactions, and argue that burden levels are often political choices rather than accidents. These sources ground the claim that outsider disadvantage in digital public services is a serious, documented phenomenon. They focus more on claimant resources and state-citizen costs than on a builder-facing requirement that outsider friction gate automation before launch.

Public-interest AI and accountable automation. ServeIT / Serve-AI clinic instincts (human review, honest handoffs, accessibility before automation) rhyme with outsider friction as a ship/no-ship input [5, 4]. Policy vocabulary about public-sector automated decision-making is useful for reading discourse; it rarely supplies a one-pager students can run before shipping clinic or civic automation. This note’s artifact targets that operational gap.

3 Approach

Method now: reflective practice and pattern extraction from lived systems use during one semester abroad, recorded as public-safe notes (no private administrative detail). Exploration versus confirmation are kept separate in the research log; confirmation experiments are planned, not run. Primary sources for this writeup are the repository essays on systems friction abroad, language and civic access, AI ethics in public services, and the Madrid-to-ServeIT practice bridge, plus the locked claim package.

The primary artifact is an Outsider Impact Review one-pager. Before shipping a civic flow or adding automation, the reviewer records the system and primary task, names an outsider persona, and marks pass / fail / n/a with a one-line note on six checks: assumed local context; unwritten sequencing; language as access (completable interface, not PDF-only “support”); hours and channels with a usable human path and case identity; recoverable errors that name the fix; automation risk (whether AI would amplify any of the above). Required-before-ship items include plain-language / language-

path fixes, sequencing stated in the UI, human override, error copy that names the fix, and deferring automation until the access baseline works. Finished reviews must copy a limitation note: single reviewer; persona-based; not a controlled study.

Practice bridge only: ServeIT / Serve-AI clinic work is a rhyme, not a relicensing of those materials. Serve-AI project materials are CC BY-SA 4.0 with project credit; this Madrid writeup and repository remain independent personal notes under MIT.

3.1 Design Principles

- Plain-language and completable language paths for the primary task (not PDF-only “support”).
- Sequencing stated in the interface; no hidden prerequisite steps.
- Recoverable errors that name the fix and preserve state.
- Human escalation with case identity when the digital path fails.
- No assumed prior belonging: design for the first-time outsider, then add automation only if the access baseline works.

4 Findings

Patterns below appeared repeatedly in lived use of civic, university, commercial, and transit/utility systems in Madrid. They suggest design failure modes; they do not prove causation beyond this observer’s semester. Recording method: public-safe reflective notes and essay outlines in madrid-ai-ethics. Observation was organized by layer (civic / government-facing; university; commercial apps; transit and utilities), asking who the form is written for and what breaks when the user is new.

1. Assumed local context. Forms and apps expect address history, bank relationships, or national ID workflows a newcomer has never used.
2. Unwritten sequencing. The flow works only if step B happened before step A, but the interface does not say so.
3. Language as access control. Spanish-only defaults; “English support” that is a PDF rather than a completable interface; language choice functions like a login wall for task completion. Mixed-language households also need structural lang marking rather than a single page default.
4. Hours and channels that do not match outsider life. In-person only, narrow windows, little or no status tracking after submit, especially hard for students and new arrivals. When phone support becomes the “real” interface, app failure equals access failure.

5. Error messages that blame the user. “Invalid document” instead of naming the fix (for example a different scan format); recoverable error UX is missing when stakes are highest.

Civic apps differ from consumer apps in a material way: the user did not choose the workflow and is trying not to miss a deadline. That raises the cost of missing progress visibility, wiping form state on one bad field, or trapping the user in a chatbot loop without a reopen path.

Same failure modes show up in ServeIT clinic observation: tools built for the maintainer’s mental model, not the visitor’s first visit. Evaluation bar for later work (not yet run): apply Outsider Impact Review and a parallel happy-path / WCAG-style pass to two public civic flows; if the outsider pass finds nothing the happy-path pass misses, the “first-class requirement” half of the claim is in serious trouble for this artifact. No baselines, sample sizes, or percentage improvements are claimed here.

5 Discussion and Limitations

Limitations are part of the claim package. Single observer: selection and confirmation bias are real. One city and one semester (Madrid, Spring 2026): other municipalities and seasons may differ. Reflective method: pattern extraction from lived use, not lab measurement. Public-safe scope omits private details that might strengthen or nuance a finding. Related-work deep reading is still owned by the author beyond the citations above; a seed list in the source repository is not a finished survey. This note does not claim legal or policy expertise (EU coursework supplied vocabulary for reading public discourse only), statistical generalizability, or that automation always harms outsiders. Poorly scoped automation can amplify gaps; that is a design risk, not a theorem. It is not IRB, clinical, or health-system research. Affiliation lines name where the author studies; they do not mean Indiana University, ServeIT Clinic, or any lab endorses or owns this content. Readers should not treat this PDF as a published conference paper or as official IU research.

6 Conclusion

Outsiders hit civic digital systems where design assumes belonging. This independent research note locks that friction as a first-class design concern, documents five public-safe patterns from Madrid reflective practice, and offers an Outsider Impact Review template for builders before automation. The next empirical step is the dual-review kill test, not invented metrics. Until that runs, the contribution is the claim, the patterns, the artifact, and the limits.

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References

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